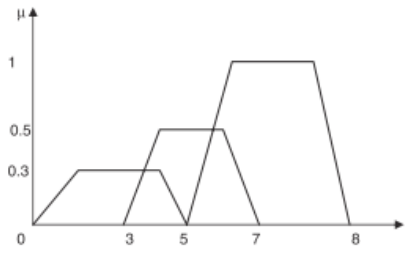
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER B. TECH (ELECTRONICS AND INSTRUMENTATION)
PROCTORED ONLINE END SEMESTER EXAMINATION Dec. 21/Jan. 22

SUBJECT: Neural Network and Fuzzy Logic (ICE 4014)

TIME: 2-5 **DATE: 17-12-2021** **MAX MARKS 20**

Note: Answer All questions.

1	A	Implement XOR function using McCulloch-Pitts neuron by considering binary data	3
	B	Implement NOR function using perceptron network for bipolar inputs and targets	3
	C	Construct a Kohonen self-organizing map to cluster the four given vectors [0 0 1 1], [1 0 0 0], [0 1 1 0] and [0 0 0 1]. The number of clusters to be formed is two. Assume an initial learning rate of 0.5.	4
2	A	Explain in detail the inference method adopted for assigning membership values.	3
	B	What is defuzzification? List different defuzzification methods. Find the defuzzified value using centroid method for the figure shown in Fig.Q2B.  Fig Q2B	4
	C	What is fuzzy proposition? Give an example for the following propositional principles • Fuzzy truth qualification • Fuzzy possibility qualification • Fuzzy probability qualification • Fuzzy usuality qualification	3